BLUE HILLS REGIONAL TECHNICAL SCHOOL
CANTON, MASSACHUSETTS

Engineering Technology

Design it. Build it. Test it. Improve it.

GRADES 9–12 · ROOM E-126 · MR. FRANK & MR. DRYER

What your student would actually do, what they leave with, and how choosing us works.

So, what is it?

Engineering Technology is the shop where students design things and then make them. A part gets drawn in professional CAD software, then printed, cut, machined or wired, and then measured against what it was supposed to do. That last step is what makes it engineering: every project ends with data, and the data is what tells a student which change to make next. Analyse, adjust, improve, measure again — that loop is the whole method.

It is not one job, either. Someone designs the part; someone works out whether it will hold; someone writes the code that moves it; someone schedules the build so it finishes on time. All of that is engineering, and it suits very different people — which is why the shop is organised around seven pathways rather than one curriculum.

Who teaches it	Mr. Frank and Mr. Dryer. Both are in the shop with all four years; Mr. Dryer delivers Grades 9 and 10, Mr. Frank Grades 11 and 12.
Where	Room E-126, plus the Makerspace next door.
How the week runs	One week in the shop — all five days, all day — then one week in academics. About twenty-five full shop days a term.
Who it suits	Students who would rather find out than be told, and who are willing to keep adjusting something in front of other people until it works.

The seven pathways

Engineering is an enormous field, and no class can stay in one part of it for long. So alongside the regular curriculum every student picks a pathway each term and goes deep on it — the same one every term to specialise, or a different one each term to come out broad. Both are normal, and changing your mind is the point rather than a problem.

PATHWAY	WHAT A TERM IN IT LOOKS LIKE
Industrial Design	Form, ergonomics, and designing something for the machine that has to make it.
Architecture & Civil	Structures, sites, drainage, and the construction drawings that go with them.
Mechanical	Forces, motion, materials — and testing a thing until it fails, on purpose.
Electrical	Circuits, power, and measuring what is actually happening rather than assuming.
Software	Logic and code that controls something physical.

Automation & Robotics	All three of the above at once. The pathway that pulls the others together.
Project Management	Scope, schedule, budget, and getting it finished. For students who want to run the project.

Things students have actually built

Not a wish list — these are real briefs that have been set, built and marked in this shop.

- **A pullback car, redesigned for a measurable improvement.** Baseline test it, find the real limitation, build the change, then prove the difference with data.
- **A take-apart toy car, reverse engineered.** Every part measured with calipers, modelled individually, then reassembled in CAD with working joints.
- **A tiny house, and an accessory dwelling unit.** Real constraints, real drawing sets, and a model at the end.
- **A city section designed around transit stops**, using the same site-analysis software urban planners use.
- **A speaker, designed and built** — enclosure, components, and a listening test that either works or does not.
- **A robot built and programmed to fight in a ring**, then an experiment sheet proving what it can and cannot push.
- **A senior capstone** the student chooses, plans, builds, tests and then defends in front of the room.

The four years

Grade 9 Engineering I	A half year, Terms 3 and 4, after exploratory. First time making something on a screen that could be made for real — CAD, the design process, drawing so somebody else can build from it, and learning to be safe in a room with machines in it.
Grade 10 Engineering II	The first full year. Documentation goes up a level, quality and testing get introduced, and **every student earns their OSHA 10 card** — which stays with them after school.
Grade 11 Engineering III	The widest year. Projects run all four terms and cover every pathway — speakers, houses, robots, circuits, architecture — before a junior capstone at the end.
Grade 12 Engineering IV	Two terms of short, sharp briefs, then the Senior Capstone: one project the student chooses and runs themselves, ending in a portfolio and a defence.

What they leave with

- **An OSHA 10 card**, earned in Grade 10, kept for life.
- **Industry certifications** in the software and equipment they have actually used — Autodesk certification is proctored here in the building, and the robotics, automation and 3D printing platforms each carry their own credential ladder.
- **Four years of documented work** — drawings, models, test data and write-ups, kept as a portfolio rather than handed in and forgotten.
- **The habit of writing it down.** Half the grade in this shop is how a student works, not what they make: safety, initiative, preparation, teamwork and professionalism are weighed as heavily as the project itself. That is deliberate, and it is how they would be judged in a real shop.

And afterwards?

Most students who leave this shop go on to college, and they go with four years of CAD, documentation and project work already behind them. Others decide on a more direct route into the field. The program is built so that both stay open — what a student does with it is their call, and we would rather they arrive at that decision having tried several kinds of engineering than having been told which one to want.

How choosing a shop works

Grade 9 students do not have to decide on day one, and they get more than a glance at each option.

- Two **mini exploratory days**, one in each of the first two terms, which between them show every one of the school's eighteen shops.
- From each of those days a student picks four or five shops to spend **a full week in** — nine week-long visits in all.
- Then they choose, and join their shop for Terms 3 and 4.

If your student spends their week with us, they are not sitting at the back watching. They are in the room with upperclassmen, working. From the day they walk in, it is their shop.

Safety, plainly

This is a room with a laser cutter, a CNC machine, power tools, heated printers and a collaborative robot in it. Nothing gets switched on by a student who has not been trained and authorised on that specific machine. Eye protection is required in the Makerspace. Closed-toe shoes always — no sandals, no Crocs. Nobody uses power tools alone. Every injury gets reported, however small.

Students wear shop attire carrying the Engineering Technology logo, ordered through the school store. It is not about looking smart; it is about being dressed for the room.

Come and talk to us. Mr. Frank and Mr. Dryer are both in E-126, and the best thing you can do is ask a student in the room what they are building and why they chose it. They will tell you more in two minutes than this sheet manages in three pages.